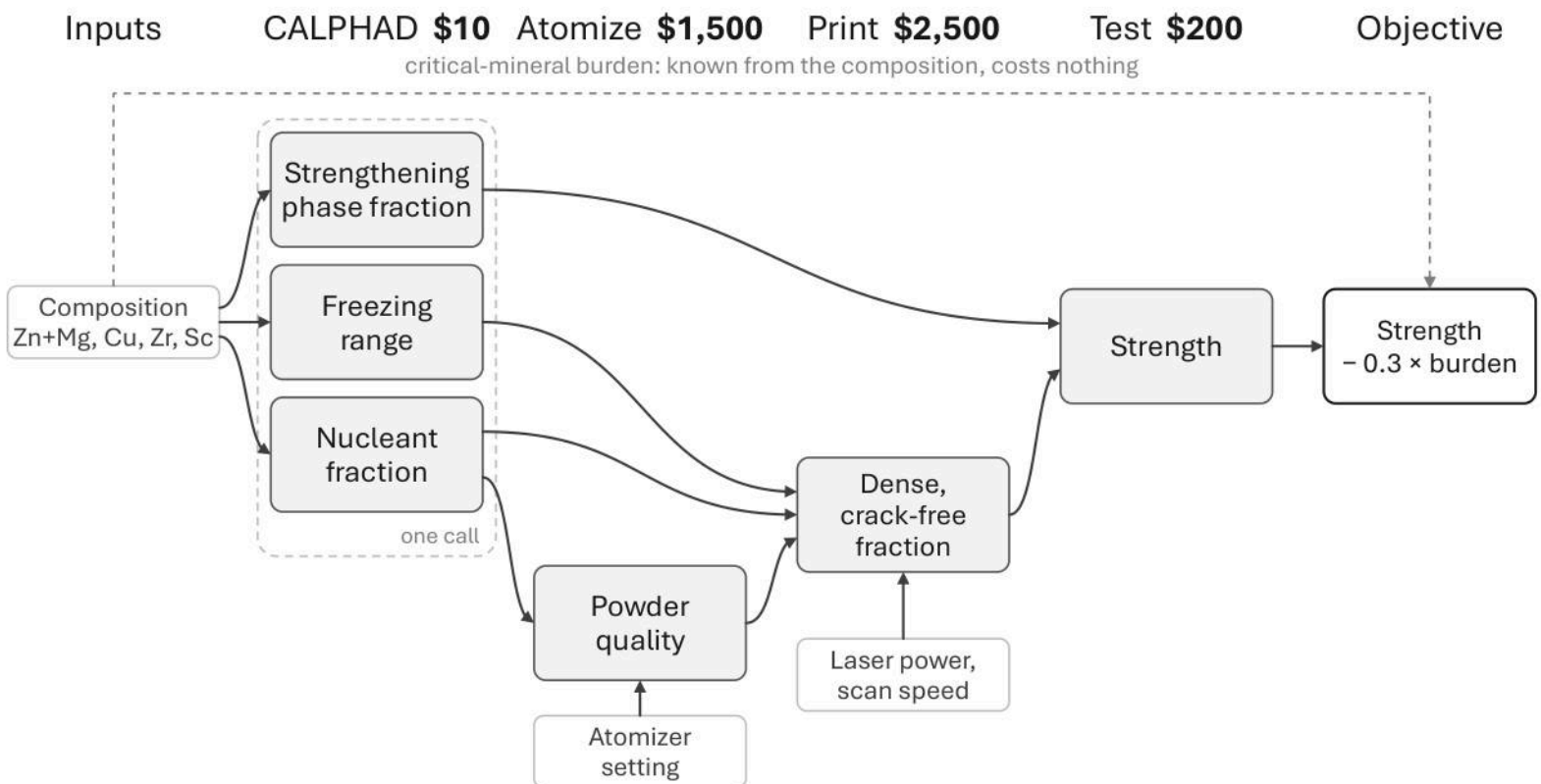
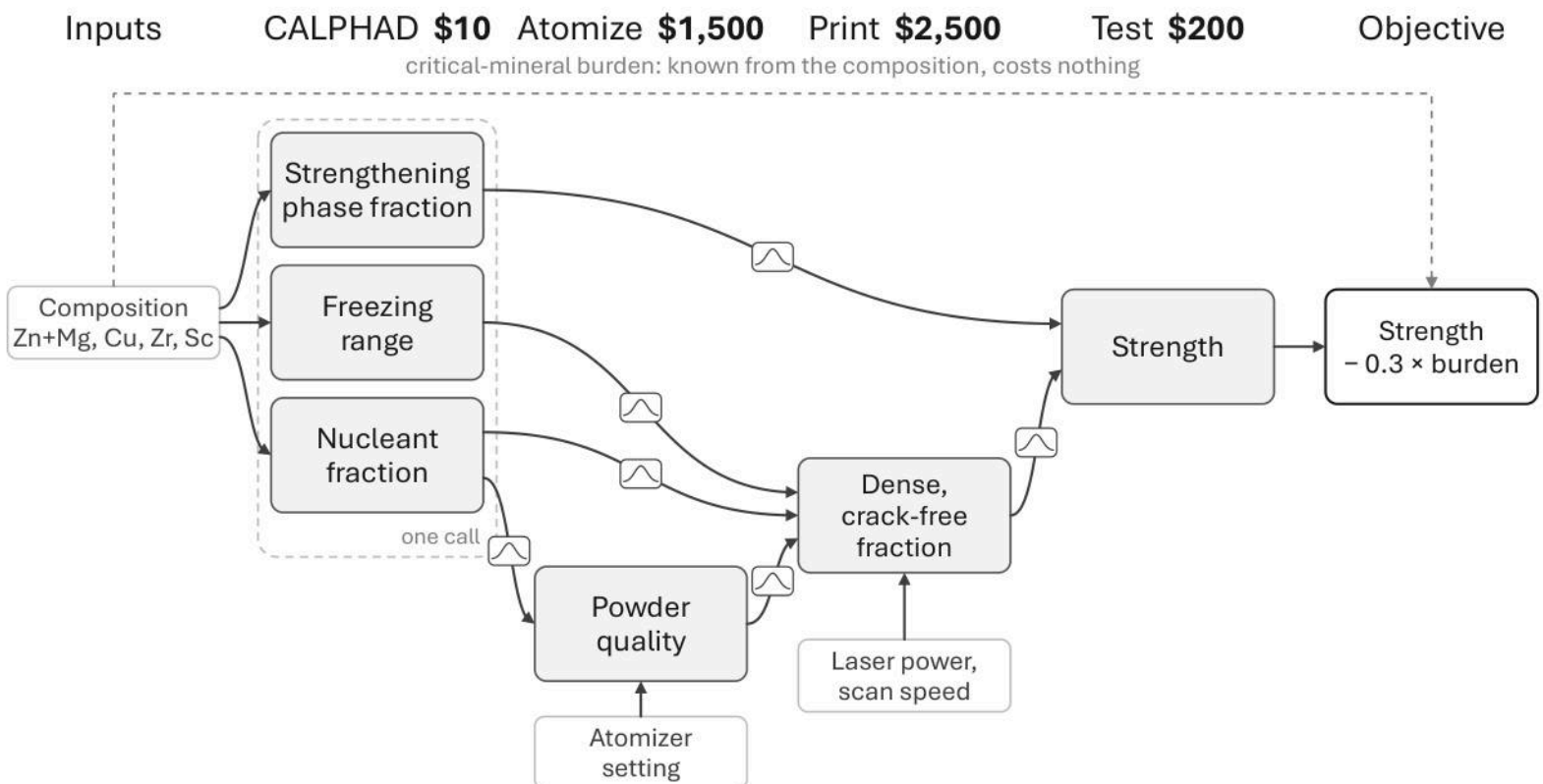
The benchmark is a network of six surrogate nodes in four priced stages

each node is a Gaussian process whose output feeds its children



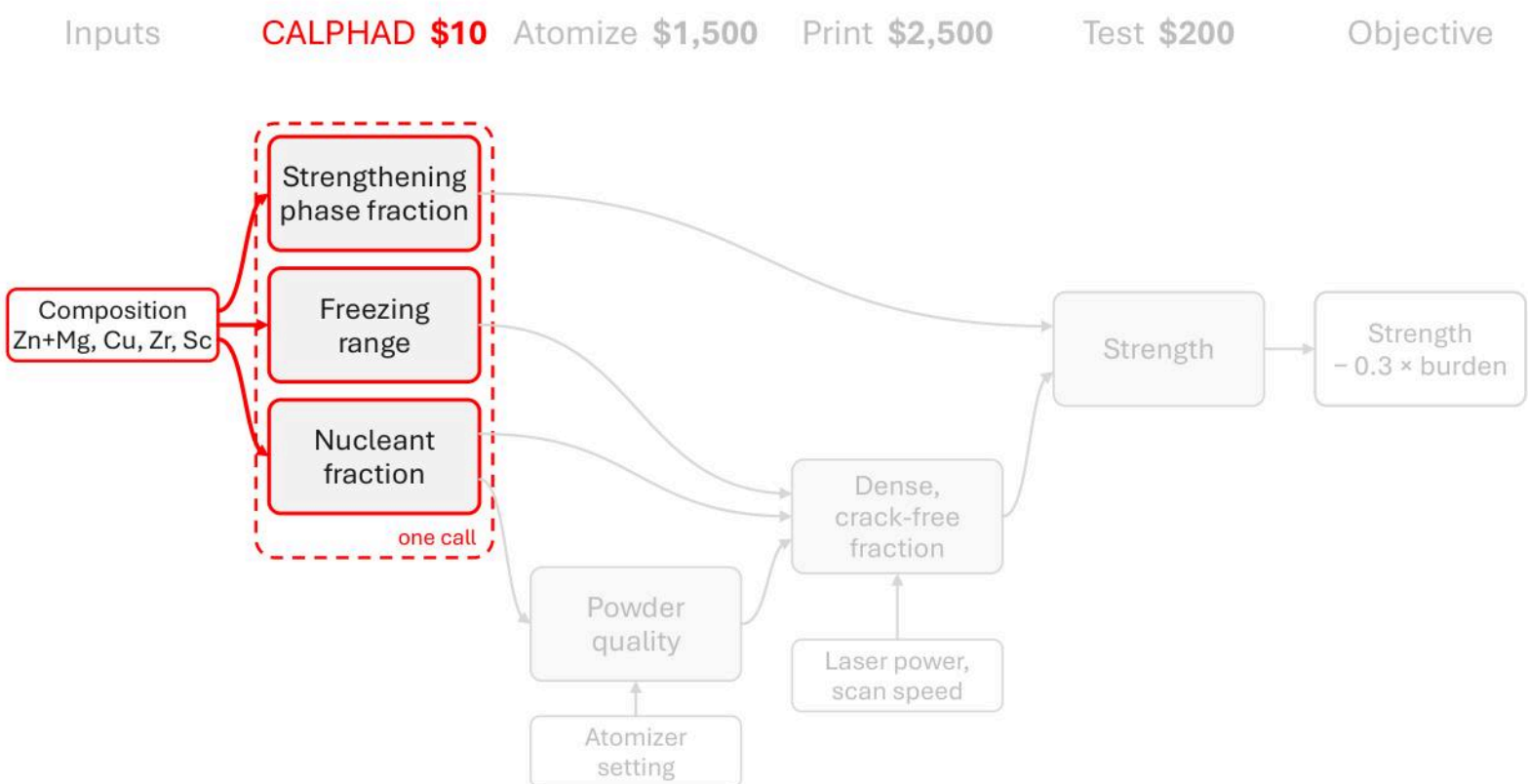
One Gaussian process per node; each child sees its parents' outputs

an observation anywhere updates everything downstream



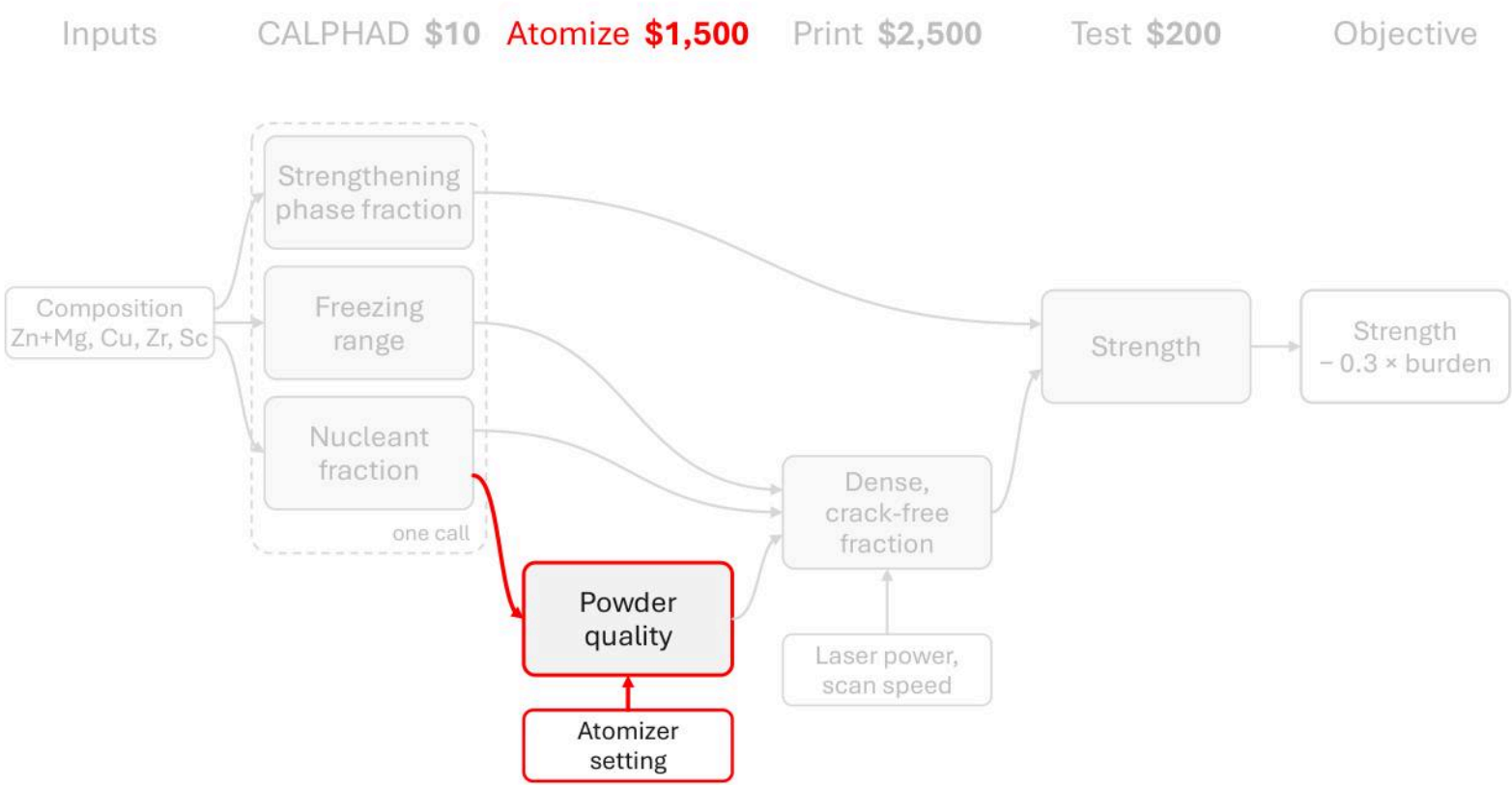
Simulate one composition for $10

one call returns all three CALPHAD outputs; nothing has to exist first



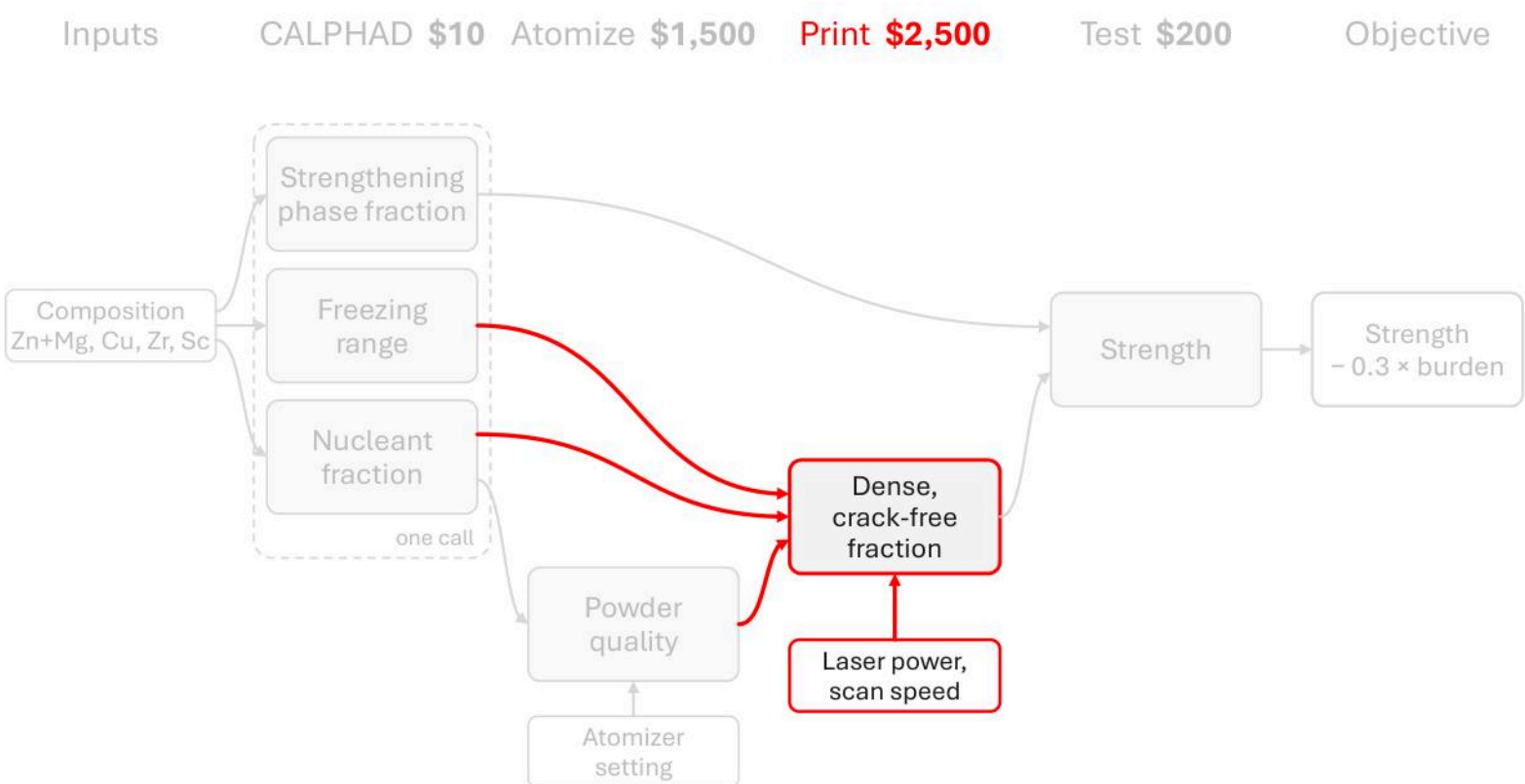
Atomize one powder lot for $1,500

needs the alloy's CALPHAD result; the lot can be printed many times

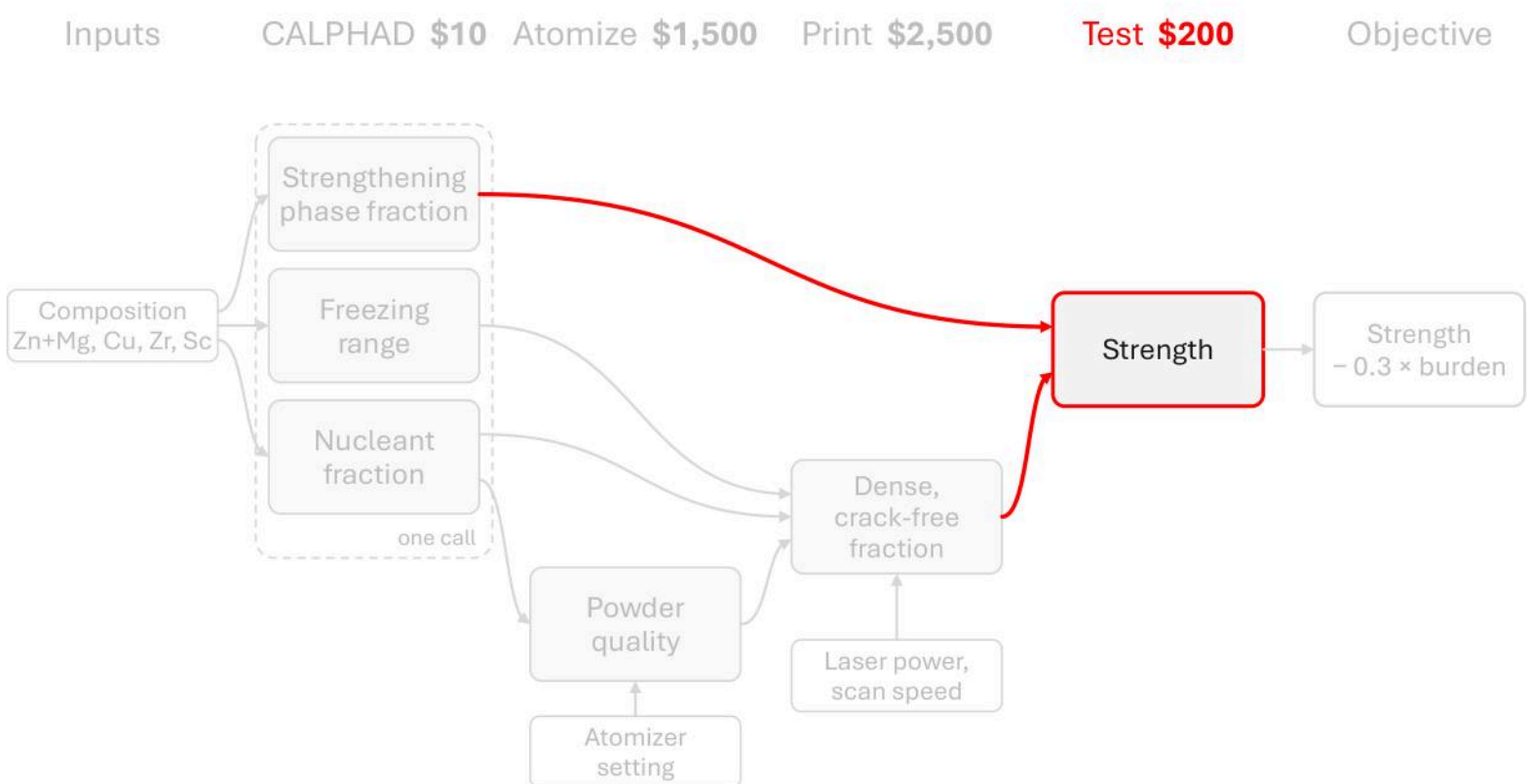


Print one build for $2,500

from a powder that exists, with one power and speed

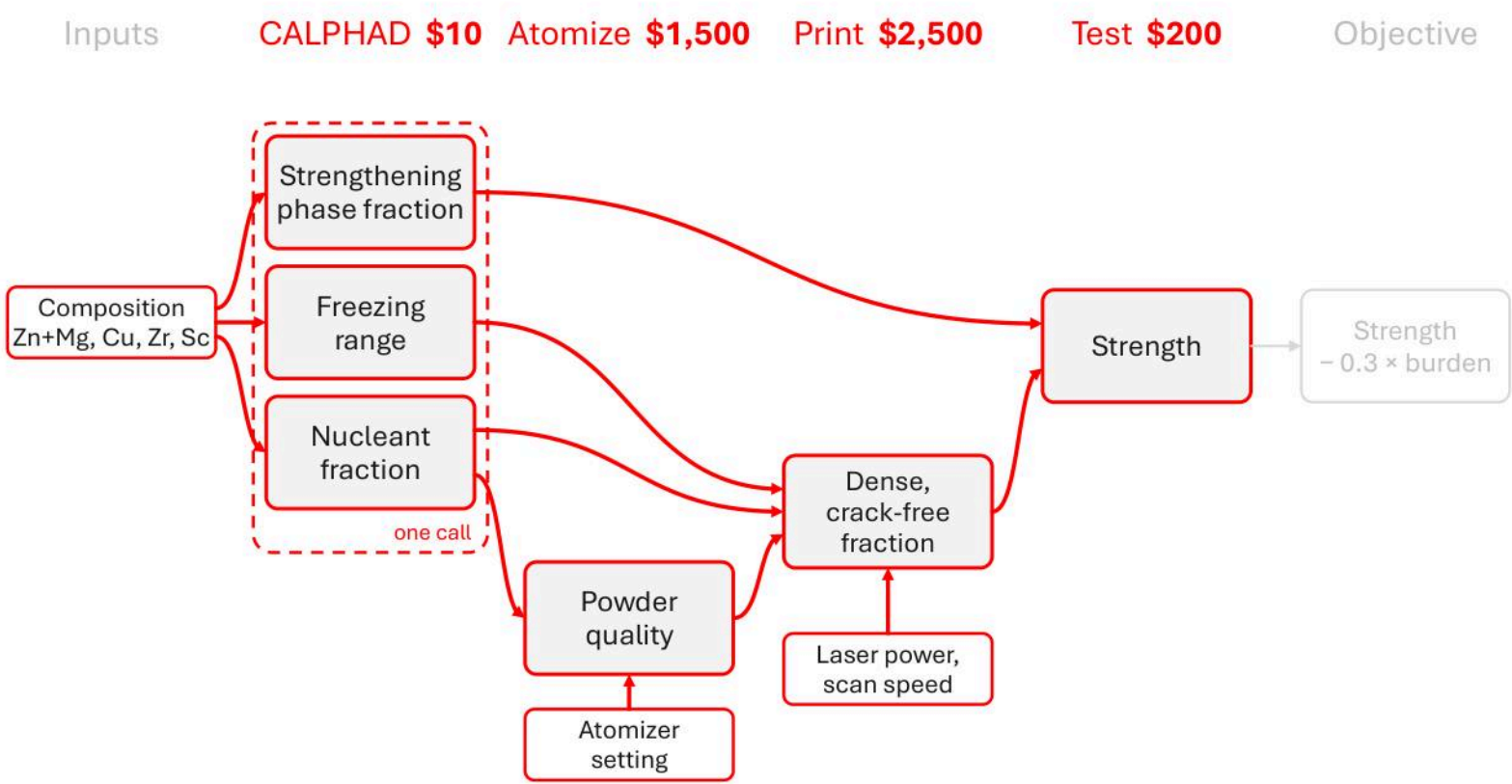


Test one bar for $200
from a build that exists



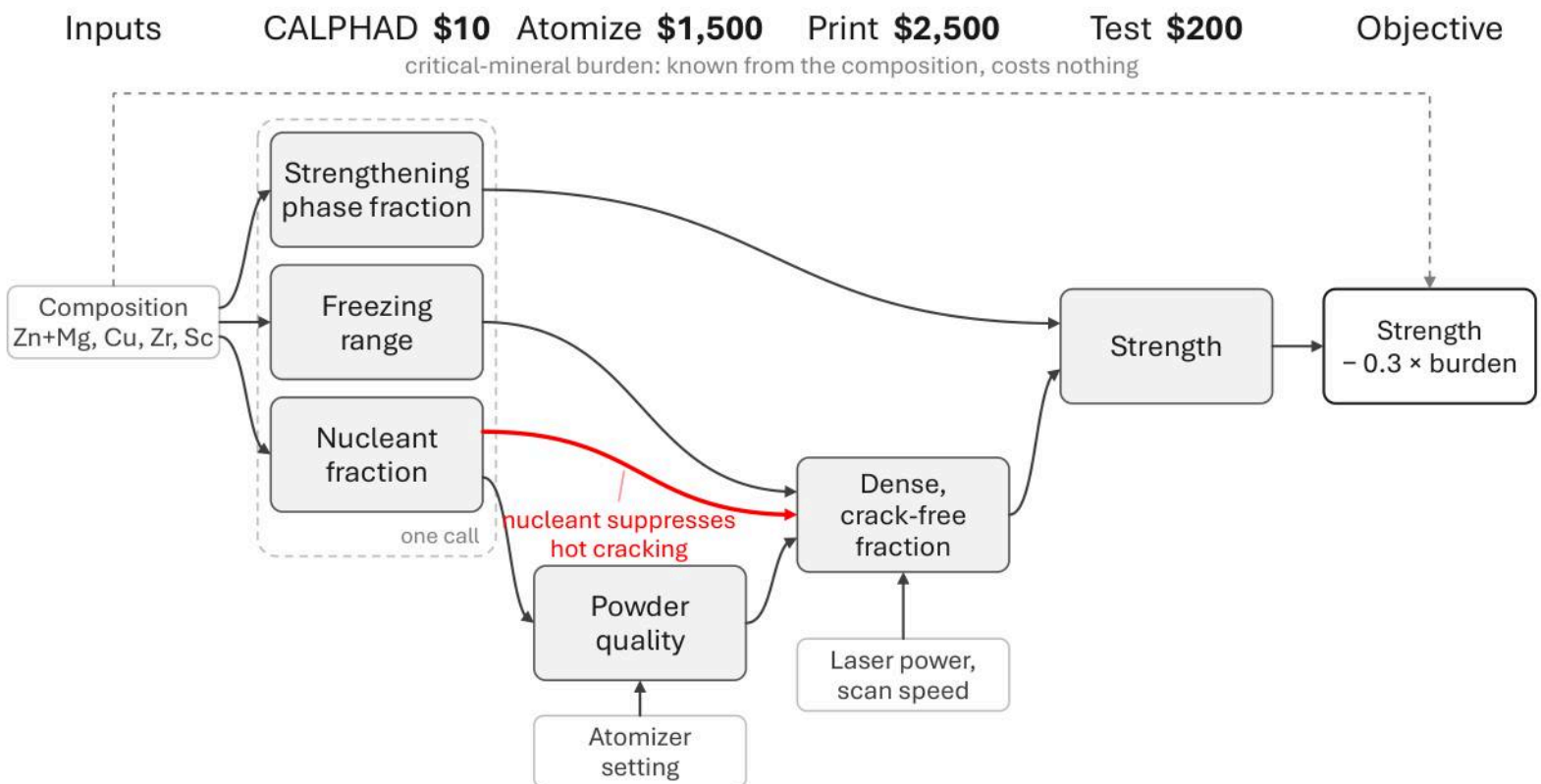
Or run the whole chain for $4,210

what standard BO, random search, tiered screening and full-pipeline BOFN buy per design



The effect a CALPHAD screen cannot see

Zr and Sc nucleants suppress hot cracking in the print; the screen ranks alloys on strength and freezing range alone



pKGFN buys one node at a time, by knowledge gradient per dollar

the full-pipeline policies buy the whole chain

